

Matthew McCaughan

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SUMMARY

Software Engineer with experience developing machine learning applications, AI-integrated systems, and database systems. Proficient in Python, Java, C++, and SQL. Projects include an interactive neural network-based digit recognition system, an AI chatbot leveraging RAG with live data injection, and a Spring Boot REST API for university database management.

EDUCATION

Rutgers University

Bachelor of Science in Computer Science

New Brunswick, NJ

Aug. 2021 – May 2025

PROJECTS

University Management REST API | [Repo](#) | *Spring Boot, Spring Data JPA, MySQL, Docker*

- Modernized a legacy JDBC command-line application into a full REST API using Spring Boot and Spring Data JPA, implementing a layered architecture (controller/service/repository/DTO/mapper) across 4 core domains (students, courses, departments, enrollments)
- Built business logic layer handling GPA calculation, transcript generation, enrollment capacity limits, and duplicate-enrollment prevention with custom exception handling mapped to appropriate HTTP status codes; containerized the full stack with Docker Compose for one-command deployment

Interactive Handwritten Digit Recognition | [Repo](#) | *Python*

- Designed and implemented an end-to-end machine learning application that trains and deploys a LeNet-5 convolutional neural network for handwritten digit classification on the MNIST dataset (~99% accuracy)
- Built an image preprocessing pipeline using OpenCV (thresholding, centering, normalization) to convert user-drawn input into model-compatible tensors
- Structured the project as a modular ML system including model training, evaluation, confusion matrix analysis, checkpointing, and an integrated user interface; enhanced the original 1998 LeNet-5 architecture with modern deep learning techniques to improve accuracy and training stability

Interactive Portfolio & Resume ChatBot | [Live Site](#) | *Gemini API, GitHub REST API, React, TypeScript, Vite, RAG*

- Built and deployed a personal portfolio website to GitHub Pages using React, TypeScript, and Vite
- Integrated an LLM chatbot via the Google Gemini API with a RAG pipeline injecting live GitHub data and resume context, enabling accurate conversational answers about skills, projects, and experience
- Automated deployment via GitHub Actions CI/CD, implementing secure secret management for API credentials and enabling continuous delivery for every push on GitHub Pages

ReceiptReader | [Repo](#) | *Java, Tess4J/Tesseract OCR, Swing, Maven*

- Built a Java desktop app that uses local OCR to scan receipt images and split itemized costs fairly among a group, with adaptive thresholding to handle varying photo quality
- Implemented a multi-format parser strategy chain to extract line items across different receipt layouts, paired with a Swing UI for reviewing and editing scanned results before splitting

SKILLS

Programming: Python, Java, SQL, C++, TypeScript

ML/AI: PyTorch, Scikit-learn, RAG

Data & Visualization: NumPy, Matplotlib, Pillow, OCR (Tesseract)

GUI & Apps: Android, JavaFX, Tkinter

Dev Tools: Docker, Maven, Git, GitHub Actions

AI Dev Tools: Google Antigravity, Claude Code, Agentic Programming

RELEVANT COURSEWORK

Data Structures, Algorithms, Computer Architecture, Databases, Software Methodology, Artificial Intelligence, Machine Learning, Computer Imaging & Multimedia